The Invariance Principle — problem 1.4

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We start with the state (a, b) where a, b are positive integers. To this initial state we apply the following algorithm :

$$\textbf{while } a > 0, \textbf{ do if } a < b \textbf{ then } (a, b) \leftarrow (2a, b - a) \textbf{ else } (a, b) \leftarrow (a - b, 2b).$$

For which starting positions does the algorithm stop ? In how many steps does it stop, if it stops ? What can you tell about periods and tails ?

The same questions, when a, b are positive reals.